

Hydroxy compounds

A Level Chemistry

Alcohols with acyl chlorides

At A Level you meet one more reaction of an **alcohol** 醇: it reacts with an **acyl chloride** 酰氯 to make an **ester** 酯. This works faster and more completely than using a carboxylic acid. For example, ethanol and ethanoyl chloride give ethyl ethanoate, plus fumes of HCl.

Phenol

Phenol 苯酚 has an -OH group joined directly to a **benzene** 苯 ring. This changes the chemistry of both the -OH group and the ring.

Making phenol

Cool **phenylamine** 苯胺 with NaNO_2 and dilute acid below 10°C to make a **diazonium salt** 重氮盐. Warming this salt with water then gives phenol (and nitrogen gas).

Reactions of phenol

- with a base such as NaOH(aq) : phenol reacts to give sodium phenoxide and water. (Ordinary alcohols do **not** react with NaOH —this shows phenol is more acidic.)
- with sodium metal: gives sodium phenoxide and hydrogen.
- with a diazonium salt in NaOH(aq) : forms a coloured **azo compound** 偶氮化合物 (used in dyes).
- **nitration** 硝化 with **dilute** HNO_3 at room temperature: gives a mixture of 2-nitrophenol and 4-nitrophenol.
- **bromination** 溴化 with bromine water at room temperature (no catalyst): gives a white precipitate of 2,4,6-tribromophenol.

Acidity of phenol

Phenol loses its H^+ to form a phenoxide ion. This ion is stabilised because its negative charge is spread (delocalised) into the ring. So phenol's **acidity** 酸性 is higher than that of water or ethanol:



(Phenol is still a weak acid —weaker than a carboxylic acid.)

Why the conditions are milder than for benzene

A lone pair on the phenol oxygen is partly **delocalised** 离域 into the ring. This makes the ring more electron-rich, so it attracts electrophiles more strongly. Phenol therefore reacts faster and under much milder conditions than benzene —no catalyst is needed, and dilute reagents work at room temperature.

The -OH group directs new substituents to the 2-, 4- and 6-positions. The same ideas apply to other phenolic compounds, such as naphthol.